

An advanced microscopy for characterization and visualization of *M. phaseolina* in chickpea roots

1. Materials

1.1 *Tissue samples*

- Control and infected root samples.

1.2 *Reagents*

- Fixative solutions: 10% neutral buffered formalin prepared in 1X Phosphate-Buffered Saline (PBS, pH 7.4); 2.5% glutaraldehyde prepared in 0.1 M sodium phosphate buffer (pH 7.2).
- Clearing and neutralization agents: 10% and 20% Potassium Hydroxide (KOH) solutions; 1 M Tris-HCl (pH 8.0 or pH 7.0); 50 mM Tris-HCl (pH 8.0 or pH 7.0).
- Embedding material: 4% high-strength agar or agarose.
- Fluorophore stains: Wheat Germ Agglutinin conjugated to Fluorescein Isothiocyanate (WGA-FITC, 50 mg/mL) (Sigma - L4895; Sigma-Aldrich Chemicals Private Limited, Bangalore, India); Calcofluor White 2MR (CFW 2MR) (Sigma - 18909; Sigma-Aldrich Chemicals Private Limited).
- Graded acetone solutions (30%, 50%, 70%, 80%, 90%, 95%, and 100% anhydrous acetone).
- Reverse Osmosis (RO) water and high-purity Milli-Q water for sample washing, section mounting, and solution preparation.

1.3 *Instrumentation and Software*

- Microscopes: Nikon SMZ25 stereo fluorescence microscope; Leica TCS SP5 or SP8 confocal laser scanning microscope systems; Scanning electron microscope (Carl-Zeiss-Strasse, Oberkochen, Germany).
- Vacuum infiltrator; Rotary shaker; Vibrating blade microtome (Leica VT1000S); Leica EM CPD300 critical point dryer.

- Quantification Software: GIMP (GNU Image Manipulation Program, version 2.10.32) configured for Microsoft Windows.

2. Methods

2.1 Confocal laser scanning microscopy (CLSM)

1. Take fresh or stressed root samples and fix them in 10% neutral buffered formalin followed by vacuum infiltration at -0.08 MPa for 30 minutes.
2. Keep the fixed samples at room temperature until microscopic observations.
3. Observe the initial tissue features under a Nikon SMZ25 stereo fluorescence microscope.
4. For confocal laser scanning microscopic analysis, wash the fixed samples with RO water thrice for 10 minutes each time.
5. Cast the washed root samples into 4% agar or agarose blocks to stabilize their structure.
6. Cut 100 μ m thick transverse sections (TS) using a Leica VT1000S vibrating blade microtome.
7. Transfer the sections into a treatment well and incubate them in 10% KOH for 8 hours in a rotary shaker at 50 rpm (or 20% KOH for 4 hours) to clear the host cytoplasmic content.
8. Replace the KOH solution with 1M Tris-HCl (pH 8.0) and incubate for 1 minute.
9. Subsequently replace the 1M Tris-HCl with 50 mM Tris-HCl (pH 8.0) to maintain the sections and equilibrate the tissue pH.
10. Transfer the cleared sections carefully onto a clean glass slide using a fine artist's brush.
11. Add 10 μ L of WGA-FITC stain (50 mg/mL) and Calcofluor White 2MR (CFW 2MR) stains directly over the transverse sections placed on the microscope slide, followed by the placement of a coverslip.
12. Incubate the slide assembly in the dark for 1 minute (see Note 1).

13. Wash the stained sections by gently flushing them with Milli-Q water to clear excess unbound fluorophore.
14. Mount the final sections and perform microscopy observations under a 20X objective lens of a confocal microscope (Leica TCS SP5/SP8) with an excitation and emission bandpass window of 495 to 515 nm for WGA-FITC and 350 to 432 nm for CFW.

2.2 Scanning electron microscopy (SEM)

1. Collect root samples from the experimental assay platform at specific tracking intervals from 1 DAI to 5 DAI.
2. Fix the collected samples immediately in 2.5% glutaraldehyde in 0.1 M sodium phosphate buffer (pH 7.2) via vacuum infiltration at -0.08 MPa for 30 minutes.
3. Dehydrate the root tissues step-by-step in a graded acetone series of 30, 50, 70, 80, 90, 95, and 100%, processing the samples twice at each concentration for 10 minutes.
4. In the end, keep the roots in fresh 100% anhydrous acetone at room temperature for 2 hours to remove all traces of water.
5. Transfer the dehydrated specimens into a Leica EM CPD300 critical point dryer to dry the tissue without collapsing cell walls.
6. Mount, sputter-coat, and capture images using a scanning electron microscope at 100X, 250X, and 500X magnifications to assess surface pathology and fungal growth structures.

2.3 Image processing and digital quantification

1. Import the captured digital microscopy images into the GIMP software layout.
2. Toggle on the Histogram dockable dialogue box via the Windows menu tab.
3. To quantify necrotic spots, select the raw brightfield/merged image window, navigate to the Value threshold channel slider, set the range window strictly from 30 to 70, and record the total pixel selection output for all biological replicates.

4. To quantify tissue fluorescence, select the standalone fluorescent image files (see Note 2).
5. Switch the Histogram selection to the green color value channel slider, set the operational threshold range window from 1 to 255, and record the cumulative green pixel count across all replicates to estimate total fungal or cellular target coverage.

3. Notes

1. WGA-FITC and CFW 2MR are highly sensitive to ambient lighting conditions. The 1-minute slide incubation and any subsequent transferring steps before mounting must be done in a dark environment or under a protective foil cover to prevent rapid photobleaching. Ensure proper bandpass optical separation during observation to prevent spectral bleed-through between channels.
2. Only raw fluorescent channel captures must be imported for GIMP Green pixel threshold evaluations. Avoid utilizing merged, multi-channel overlays or composite brightfield images for fluorescence calculations, as non-specific background structural reflections will falsely inflate the recorded pixel count.

Please cite the original article for this method:

- Irulappan, V. et al. (2022). Drought Stress Exacerbates Fungal Colonization and Endodermal Invasion and Dampens Defense Responses to Increase Dry Root Rot in Chickpea. *Molecular Plant-Microbe Interactions*, 35(7), 583-591.
- Mirchandani, R., Kandpal, M., Ranjan, A., Sinharoy, S., & Senthil-Kumar, M. (2025). Induced post-invasive defenses in the nonhost plant *Parthenium hysterophorus* L. prevent root cortical colonization by *Macrophomina phaseolina* and impart resistance to dry root rot. *Environmental and Experimental Botany*, 237, 106197.